

Tawsif Mayaz

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EDUCATION

University of Waterloo

Sept 2023 – Apr 2028

Bachelor of Applied Science, Computer Engineering

Waterloo, ON

- * Coursework: Data Structures & Algorithms, Systems Programming & Concurrency, Real-Time Operating Systems, Computer Architecture (RISC-V), Digital Hardware Systems

EXPERIENCE

Software Developer, Part-Time

May 2026 – Present

Empire Life Insurance Company

Toronto, ON

- * Delivered a **Java/Spring** API returning any record's state as of a past date, refactoring to hexagonal architecture that moved **150+** lines of persistence logic into adapters
- * Migrated a downstream service onto a new internal **API**, implementing **8 REST** integrations behind one interface and securing service-to-service calls with scoped **Auth0** grants
- * Designed end-to-end test automation spanning a TCP listener, message queue, and **REST API** across **12+** scenarios, replacing a shared consumer with per-request queues to eliminate flaky failures

Software Developer Intern

Jan 2026 – Apr 2026

Empire Life Insurance Company

Toronto, ON

- * Built an event-driven **ETL** pipeline on **Google Cloud Pub/Sub** and **SQLAlchemy** that has loaded **4,000+** partner records across **600+** files with **zero** failed imports
- * Extended the platform to run on either **SQL Server** or legacy **Sybase** from one codebase, validating a live database migration across **5** services with zero downtime
- * Provisioned buckets, topics, and scoped service accounts as reusable **Terraform** modules and shipped distroless **Docker** images via **Helm** and **Argo CD**, making every environment reproducible from version control
- * Built a **Datadog** observability dashboard on **OpenTelemetry** traces, wiring **14** monitors over imports and failures

Software Developer Intern

May 2025 – Aug 2025

Cineplex Digital Media

Waterloo, ON

- * Built a real-time spatial system in **Unity (C#)** that reconstructs retail environments and models customer touch behavior from depth-sensing data
- * Cut in-store merchandising setup time by **60%** with an interactive 3D configuration workflow backed by **SQLite** persistence
- * Scaled analytics to **10,000+** interactions per session through a **React** and **Flask** pipeline visualizing customer engagement

Full-Stack Developer

Sept 2024 – Apr 2025

Electrium Mobility

Waterloo, ON

- * Built an e-commerce platform for electric vehicle rentals with **Next.js** and **Supabase**, covering checkout and logins

Full-Stack Development Intern

Jan 2024 – Aug 2024

Mhapy

Toronto, ON

- * Shipped **10+** features on an Agile team of **7** for a patient-therapist matching platform, raising engagement **30%** with **React**
- * Developed recommendation workflows with **NestJS**, **Express**, and **PostgreSQL** that improved matching accuracy **25%**

PROJECTS

Sibyl - 1st Place & Best Use of AI/ML (Hack for Humanity) | *React, FastAPI, PostgreSQL, pgvector, SSE*

- * Won among **750+** participants building an AI auditor for corporate sustainability reports that checks every extracted claim against satellite, legal, news, and academic evidence
- * Owned the compliance report view and its backend compiler, plus a **RAG** chatbot that fans out to **7** sources in parallel over **pgvector** and full-text search, streaming cited answers over **SSE**

Rosetta - 3rd Place (HackHive) | *React, FastAPI, PostgreSQL, RAG, LLMs*

- * Built a **RAG** pipeline with **BGE embeddings**, **KeyBERT**, and cross-encoder re-ranking that embeds course-note citations into live translations as professors speak
- * Streamed transcription over **WebSockets** between a **React** frontend and **FastAPI** backend, generating LLM-driven structured notes

TECHNICAL SKILLS

Languages: Python, Java, TypeScript/JavaScript, C/C++, C#, SQL

Frameworks / Libraries: React, React Native, Next.js, Node.js, Express.js, NestJS, Spring Boot, FastAPI, Flask, Tailwind CSS

Tools: GCP (Pub/Sub, GCS), Docker, Kubernetes, Terraform, Argo CD, PostgreSQL, MSSQL, OpenTelemetry, Datadog, Git